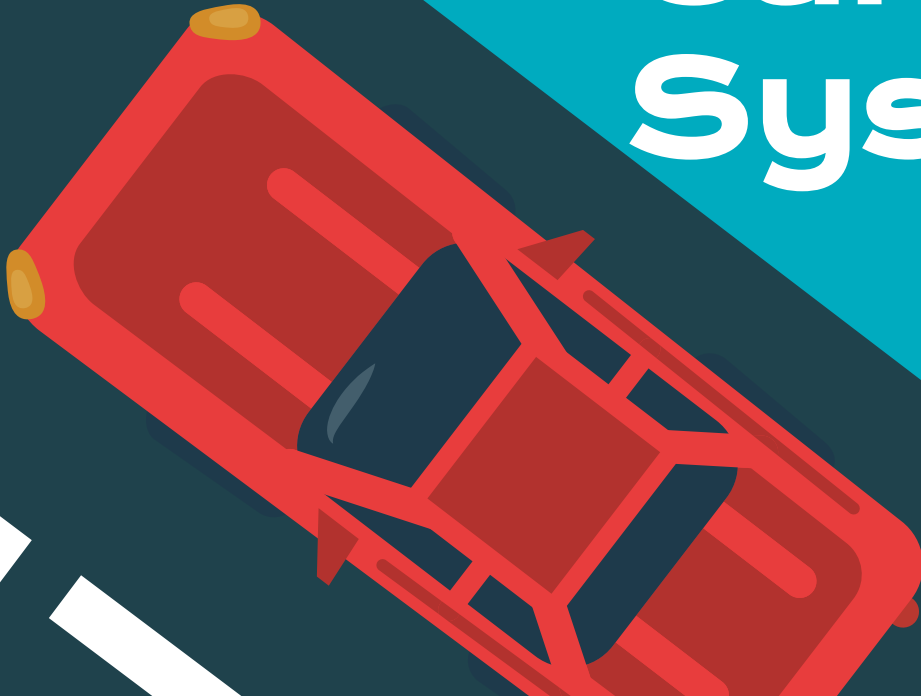


Car Rental System

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Description

The Car Rental System is an application whose purpose is to easily help customers rent vehicles through the use of an automated system. Customers can select from a variety of vehicles and book them and make payments through the system and at the same time employees can track the vehicles currently rented and report to the higher ups. The idea of this Car Rental System is to make an automated online system that takes care of the whole process of renting a car starting from reserving a car to making payments to returning the car. This system will be more commonly used compared to manual car rentals as it eliminates the challenges of manual car rentals.



Classes

CUSTOMER

BUYER

RENTAL

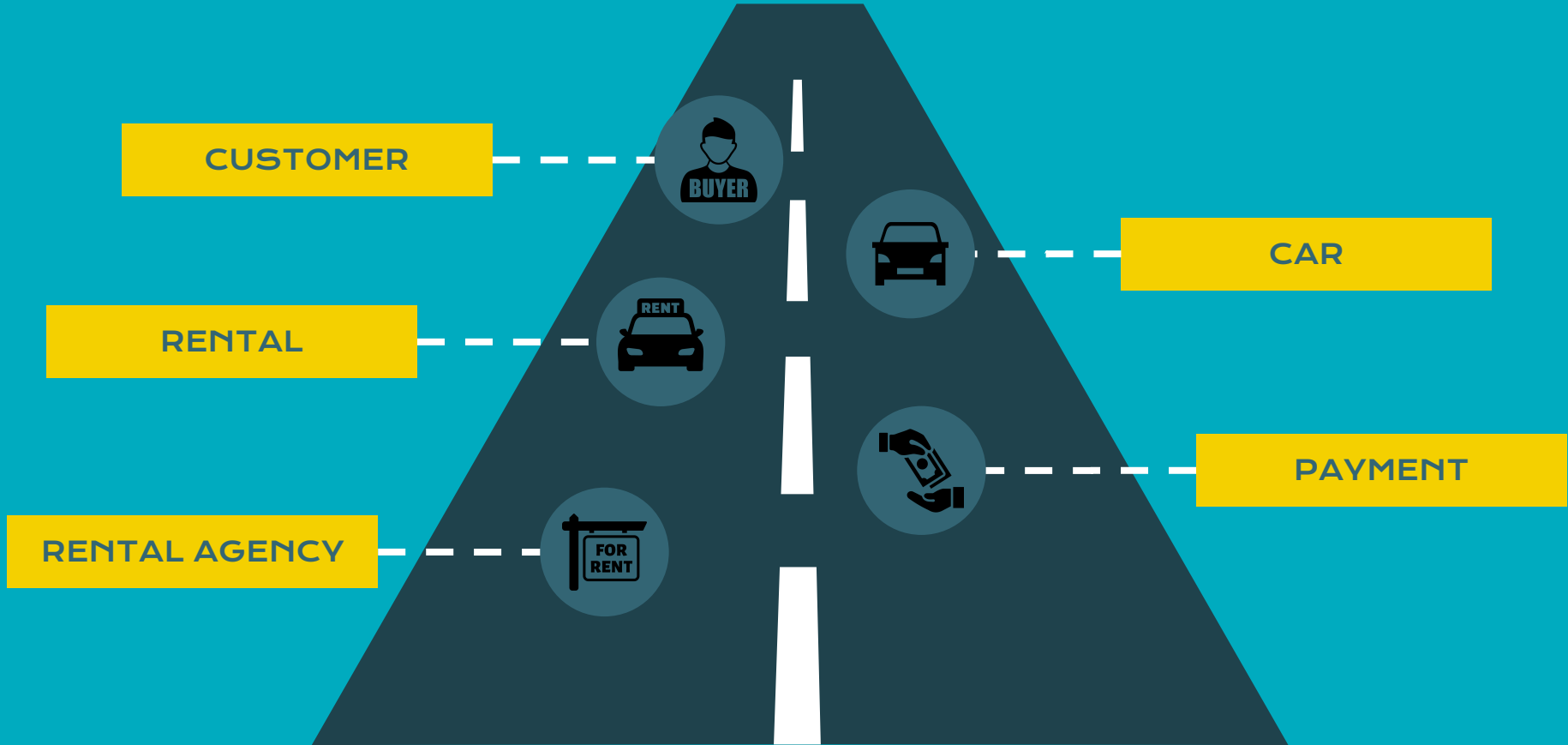
RENT

CAR

RENTAL AGENCY

FOR
RENT

PAYMENT



Each class and its responsibility

- **Customer :** The customer class represents a person who wants to rent a car from us. It stores all the customers personal information needed to identify the customer and process his rental. Every rental transaction is linked back to the customer object.
- **Car:** The car class represents a vehicle available for rent in the agencies collection. It keeps track of the cars details and whether its available or already rented out. The cost of rental varies depending on the type of car.
- **Rental:** The rental class is the core transaction of the system. It connects a customer to a car and records the start date and end date and automatically calculates the total based on the duration the car was rented. It also tracks whether the rental is still active or closed.
- **Payment:** The payment class takes care of the financial aspect of the system. When a customer confirms the rental of a certain vehicle they complete the payment by using their preferred payment method. The transaction is processed in the payment class and a receipt will be generated for the customer.
- **Rental Agency :** This class controls the entire system. All the main functions of the system from cars to customers to payments happens in this class.

Class Relationships

Our system will use two types of relationships between classes: Inheritance and composition

We will have two inheritance relationships:

1. Person → Customer: The Customer class will inherit from Person. This means Customer will automatically get the attributes of person like name, ID and phone number from person and then it will have its own attributes like drivers license number and others.
2. Car → Sedan, SUV, Truck: We will have sub classes called Sedan, SUV and Truck and they will all inherit from Car. They share the same basic car attributes but each one calculates its price differently depending on the type of car they rent.

Class Relationships

Our system will have 6 composition relationships:

1. RentalAgency has Cars: The agency will own and manage all the cars in our collection
2. RentalAgency has Customers: The agency will store and manage all the registered customers
3. RentalAgency has Rentals: The agency creates and tracks all the rental transactions
4. Rental has a Car: Every rental will be linked to one specific car being rented
5. Rental has a Customer: Every rental is linked to a customers who made the reservation
6. Rental has a Payment: Every rental will automatically generate one payment object to handle the transaction